ENHANCING INCLUSIVE EDUCATION IN SRI LANKA THROUG A PERSONALIZED MULTIMODEL LEARNING PLATFORM FOR CHILDREN WITH DOWNSYNDROME

Dinula Chamadith Fonseka

IT22341518

BSc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

VOCATIONAL TRAINING MODULE FOR THE DOWN SYNDROME STUDENTS

Dinula Chamadith Fonseka

IT22341518

BSc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

DECLARATION

I declare that this is my own work and this dissertation1 does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Signature: Date: 25 / 04 / 2026

The supervisor/s should certify the dissertation with the following declaration. The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

25/04/2026

Signature of the supervisor: Date:

(Ms. Hansika Mahaadikara)

ABSTRACT

The Career Practice Module of the Nanapiyasa Learning Platform is designed to support students with Down syndrome in developing practical, real-world vocational skills and progressing toward greater independence. The module focuses on identifying each learner's preferences, abilities, and behavioral characteristics through a structured questionnaire, enabling personalized career guidance.

The system begins with an assessment that includes rating-scale and capability-based questions evaluating social interaction, organization, communication, task engagement, memory, and teamwork. The responses are converted into quantitative scores and analyzed to generate percentage-based suitability levels for selected job roles. Using a machine learning–based recommendation approach, the system identifies the most suitable role by selecting the highest compatibility score aligned with the student's profile.

The module currently targets three vocational roles: Chef Assistant, Retail Assistant, and Cleaning Assistant. Based on the recommended role, learners participate in interactive, scenario-based activities that simulate real-world tasks. These include preparing a fruit salad and making tea (Chef Assistant), billing items and assisting customers in a retail setting (Retail Assistant), and performing basic cleaning tasks (Cleaning Assistant). Each activity is designed with step-by-step instructions, visual guidance, and repetitive practice tailored to the cognitive needs of the learners.

By integrating personalized assessment, intelligent role recommendation, and hands-on learning experiences, the Career Practice Module aims to enhance functional independence, build confidence, and prepare students for real-world work environments. This approach supports the transition toward a more self-reliant lifestyle while reducing dependency on caregivers and promoting inclusive participation in the workforce.

Keywords - Down syndrome, Job Practice Module, Personalized Learning, Machine Learning, Vocational Skill Development

ACKNOWLEDGEMENT

The authors would like to express their sincere gratitude to all individuals and institutions who contributed to the successful completion of this research study.

We extend our appreciation to the academic staff of the Sri Lanka Institute of Information Technology (SLIIT) for their continuous guidance, constructive feedback, and academic support throughout the research process. Their expertise and encouragement were invaluable in shaping the direction and quality of this work.

We also acknowledge the contribution of all participants who provided data through the Google Forms survey. The availability of real-world data from approximately 600 students significantly contributed to the reliability and validity of the proposed model.

Contents

INTRODUCTION 1
1.2 Role of MJF Charitable Foundation in the Study 2
Overview of MJF Charitable Foundation 2
Goals of MJF Charitable Foundation 3
Relevance to this research 3
1.3 Research Gap 3
1.4 Research Objectives 8
Main objective 8
Specific objectives 8
Career Practice Module 8
1.6 Data Collection and Dataset Development 8
1.7 Significance of the study 9
2. METHODOLOGY 9
2.1 Methodology Overview 9
1. Frontend Layer 10
2. Backend Layer 11
3. Data Layer 11
2.3 Data Collection Methodology 11
2.3.1 Target population 11
2.3.2 Data collection tool 11
2.3.3 Role of experts 11
2.3.4 Data Categories 12
2.3.5 Output Labeling 12
2.4 Questionnaire Design and Behavioral Mapping 12
2.4.1 Rating-Based Questions (1–5 scale) 12
2.4.2 Capability-Based Questions 12
2.4.3 Preference-Based Questions 12
2.4.4 Feature Mapping Strategy 13
2.5 Data Preprocessing 13
2.5.1 Cleaning 13
2.5.2 Normalization 13
2.5.3 Encoding 13
2.5.4 Feature Engineering 13
2.5.5 Dataset Splitting 13
2.6 Machine Learning Model for Career Recommendation 14

2.6.1 Input Features 14
2.6.2 Output Labels 14
2.6.3 Model Selection 14
2.6.4 Training Process 15
2.6.5 Prediction Process 15
2.7 Career Simulation Module 15
2.7.1 Purpose 15
2.7.2 Simulation Scenarios 15
2.7.3 Interaction Design 16
2.7.4 Task Execution Flow 16
2.7.5 Scoring System 16
2.8 Commercialization Aspects of the Product 16
2.8.1 Target Market and User Segments 17
2.8.2 Value Proposition of the System 17
2.8.3 Revenue and Sustainability Model 18
2.8.4 Competitive Advantage 18
2.8.5 Scalability and Future Expansion 18
2.8.6 Social Impact 19
2.9 Testing and Implementation 19
2.9.1 System Implementation Overview 20
2.9.2 Functional Testing 20
2.9.3 Machine Learning Model Evaluation 21
2.9.4 User Testing and Validation 21
2.9.5 Performance and Usability Testing 22
2.9.6 Deployment and System Integration 23
2.9.7 Limitations Observed During Testing 24
2.9.8 Summary of Implementation Outcomes 26
3. RESULTS AND DISCUSSION 29
3.1 Results 29
3.1.1 Overview of Experimental Results 29
3.1.2 Machine Learning Prediction Results 30
3.1.3 Simulation Performance Results 32
3.1.4 Feedback from Educators and Therapists 32
3.2 Research Findings 35
3.2.1 Behavioral Pattern Identification 35
3.2.2 Effectiveness of Questionnaire-Based Profiling 35

3.2.3 Impact of Simulation-Based Learning 35
3.2.4 Role of Expert Validation 35
3.3 Discussion 35
3.3.1 System Effectiveness 35
3.3.2 Comparison with Traditional Methods 36
3.3.3 Advantages of the Proposed System 36
3.3.4 Limitations and Challenges 36
3.3.5 Future Improvements 36
3.4 Summary of Each Student's Contribution 37
3.4.1 Data Collection and Dataset Preparation 37
3.4.2 Machine Learning Development 37
3.4.3 Frontend Development 37
3.4.4 Backend Development and Integration 37
3.4.5 Testing and Documentation 37
3.5 Overall Summary of Results and Discussion 37
CONCLUSION 39
REFERENCES 40

List of Tables

Table 1: Feature Mapping Strategy

List of Figures

Figure 1: Vocational Module System Architecture

Figure 2: Neural Network Accuracy Curve

Figure 3: Neural Network Loss Curve

Figure 4: Testing the app with students

Figure 5: Teacher's feedback regarding the project

INTRODUCTION

1.1 Background Literature

In recent years, there has been increasing global attention on inclusive education, assistive technologies, and vocational empowerment for individuals with intellectual and developmental disabilities. Among these groups, individuals with Down syndrome face unique challenges in cognitive processing, adaptive behavior, communication, and independent living skills. These challenges often limit their access to traditional education pathways and employment opportunities. However, recent studies indicate that with structured learning environments, repetition-based instruction, and visual guidance, individuals with Down syndrome can effectively develop functional and vocational skills necessary for independent living [1], [2].

Vocational training has been widely recognized as a critical factor in improving independence and overall quality of life for individuals with intellectual disabilities. Employment opportunities not only provide financial benefits but also contribute to improved self-esteem, social participation, and emotional well-being. According to recent research, structured vocational training programs significantly enhance task completion abilities, workplace adaptability, and long-term employability outcomes among neurodiverse populations [3], [4]. Furthermore, international educational frameworks emphasize the importance of inclusive vocational pathways as a means of reducing social exclusion and promoting equal opportunities in the workforce [5].

Despite these advancements, traditional vocational education systems still rely heavily on instructor-led teaching methods and physical training environments. While these approaches can be effective, they often lack flexibility and personalization. Each learner demonstrates unique cognitive abilities, behavioral traits, and learning speeds, which require adaptive teaching strategies. In many educational settings, instructors are required to manually tailor training approaches, which can lead to inconsistencies and increased workload [6]. Additionally, access to such training programs may be limited in developing regions, further restricting opportunities for individuals with disabilities.

With the rapid advancement of digital technologies, there has been a growing interest in the application of artificial intelligence in education. Machine learning techniques have been increasingly used to analyze learner behavior, predict learning outcomes, and personalize educational content [7]. These intelligent systems enable data-driven decision-making, allowing for more accurate identification of individual learning needs. In the context of special education, AI-driven systems can support adaptive learning environments that respond dynamically to user performance and preferences.

Simulation-based learning has also emerged as a powerful tool in vocational education. Simulations provide learners with the opportunity to practice real-world tasks in a safe and controlled environment. This approach is particularly beneficial for individuals with cognitive challenges, as it allows repeated practice without real-world risks. Studies have shown that

simulation-based training improves procedural understanding, task accuracy, and confidence levels among learners with intellectual disabilities [8]. Moreover, interactive simulations enhance engagement by providing visual and experiential learning opportunities, which are more effective than traditional text-based instruction.

In recent years, there has been a shift towards integrating artificial intelligence with simulation-based learning to create intelligent training systems. These systems combine data analysis, personalized recommendations, and interactive environments to deliver a comprehensive learning experience. For example, AI-driven vocational platforms can recommend suitable career paths based on behavioral data and guide learners through role-specific training scenarios. Such systems have shown promising results in improving skill acquisition and learner engagement [9].

However, despite these technological advancements, most existing systems are designed for general education or high-level professional training. There is a noticeable lack of solutions specifically tailored for individuals with Down syndrome and similar cognitive conditions. Many platforms do not adequately consider the unique learning requirements of these individuals, such as the need for simplified interfaces, repetitive instruction, and guided interaction.

Furthermore, existing vocational guidance systems often focus only on assessment or training, rather than providing a complete pipeline that includes data collection, intelligent recommendation, and practical simulation. This fragmentation limits their effectiveness and usability in real-world educational settings. There is therefore a clear need for an integrated system that combines behavioral assessment, machine learning-based recommendation, and simulation-based training into a unified framework.

The Career Practice Module proposed in this study aims to address these limitations by introducing a comprehensive, data-driven approach to vocational training for students with Down syndrome. By leveraging real-world data, machine learning algorithms, and interactive simulations, the system provides personalized career recommendations and practical training opportunities. This approach aligns with current trends in inclusive education and demonstrates the potential of intelligent systems in enhancing vocational outcomes for neurodiverse learners.

1.2 Role of MJF Charitable Foundation in the Study

A critical part of this research is the collaboration with the MJF Charitable Foundation, which played a significant role in both data collection and domain validation.

Overview of MJF Charitable Foundation

The MJF Charitable Foundation is a well-established non-governmental organization in Sri Lanka dedicated to empowering marginalized and vulnerable communities. The foundation focuses on education, healthcare, livelihood development, and disability support services. Its

mission is to create an inclusive society where individuals with disabilities can access equal opportunities for education, skill development, and employment.

The foundation operates several centers that provide specialized training programs for individuals with disabilities, including those with intellectual and developmental challenges such as Down syndrome. These centers are staffed with trained teachers, therapists, and caregivers who support learners through structured programs tailored to their abilities.

Goals of MJF Charitable Foundation

The primary goals of the foundation include:

- Promoting inclusive education for individuals with disabilities
- Enhancing vocational skills for employment readiness
- Supporting independent living and social integration
- Providing therapy and behavioral support services
- Reducing dependency on caregivers through skill development

Relevance to this research

In the context of this study, the MJF Charitable Foundation served as a key partner in:

- Facilitating access to students for data collection
- Providing expert guidance from teachers and physiotherapists
- Validating questionnaire design and classification of job roles
- Ensuring ethical and appropriate handling of student data

The involvement of the foundation ensured that the dataset used in this research is grounded in real-world observations and expert assessments, increasing the reliability and relevance of the machine learning model.

1.3 Research Gap

Despite significant advancements in assistive learning technologies and inclusive education frameworks, several critical gaps remain in the domain of vocational training for students with Down syndrome. These gaps limit the effectiveness of current systems and highlight the need for more integrated, personalized, and data-driven solutions.

Firstly, most existing systems do not provide personalized career recommendation mechanisms. While many digital learning platforms offer general skill development modules, they rarely incorporate behavioral analysis to guide career selection. In traditional systems, vocational pathways are often assigned based on broad categorizations rather than individual learner profiles. Recent research emphasizes that personalized learning systems powered by

machine learning can significantly improve learning outcomes by aligning training with individual abilities and preferences [10], [11]. However, the application of such personalized recommendation techniques in vocational training for individuals with Down syndrome remains limited.

Secondly, there is a notable lack of structured and domain-specific datasets tailored for neurodiverse vocational training. Most existing intelligent systems rely on generic datasets that do not accurately represent the behavioral, cognitive, and functional diversity of learners with intellectual disabilities. This results in reduced model accuracy and limited generalizability. Studies highlight that the effectiveness of machine learning models heavily depends on the quality and relevance of training data, particularly in specialized domains such as special education [12]. The absence of real-world, labeled datasets that capture behavioral patterns of students with Down syndrome presents a significant barrier to developing reliable recommendation systems.

Thirdly, there is insufficient integration of domain expert knowledge in system design. Teachers, therapists, and caregivers possess deep insights into learner behavior, strengths, and challenges. However, these insights are often not systematically incorporated into computational models. As a result, many data-driven systems lack contextual understanding and fail to reflect real-world educational practices. Research suggests that combining expert knowledge with machine learning models can significantly enhance system accuracy and reliability, particularly in domains requiring human-centered decision-making [13]. The lack of such integration creates a disconnect between technological solutions and practical educational needs.

Another major gap is the limited availability of interactive simulation-based training environments specifically designed for individuals with cognitive disabilities. While simulation-based learning has been widely adopted in general education and professional training, most existing systems are not tailored to the unique requirements of neurodiverse learners. These systems often lack simplified interfaces, guided instructions, and adaptive difficulty levels necessary for effective learning. Studies have shown that simulation environments designed with accessibility and cognitive support features can significantly improve engagement and skill acquisition among learners with intellectual disabilities [14]. However, such specialized systems are still scarce in vocational training contexts.

Furthermore, there is a lack of adaptive and continuous evaluation mechanisms in existing systems. Traditional vocational training methods often rely on periodic assessments rather than continuous performance tracking. This limits the ability to monitor progress and provide timely feedback. Recent advancements in educational technologies highlight the importance of real-time analytics and performance monitoring in improving learning outcomes [15]. However, these capabilities are not widely implemented in systems designed for special education.

Another critical gap lies in the absence of end-to-end integrated systems that combine all essential components of vocational training into a unified framework. Most existing solutions

address only one aspect of the process, such as assessment or training, without integrating them into a cohesive system. The key components that are often fragmented include:

- Assessment (questionnaire-based profiling)
- Recommendation (machine learning-based role selection)
- Training (simulation-based learning)
- Evaluation (performance scoring and feedback)

This fragmentation reduces system effectiveness and creates challenges in maintaining consistency across different stages of learning. Research indicates that integrated learning systems provide better outcomes by ensuring continuity and alignment between assessment, instruction, and evaluation [16].

In addition, there are significant challenges related to accessibility and usability in existing assistive technologies. Many systems are not designed with the cognitive limitations of users in mind, resulting in complex interfaces and difficult navigation. This reduces user engagement and limits the practical applicability of such systems. Studies emphasize the importance of user-centered design principles in developing accessible technologies for individuals with cognitive disabilities [17].

In the context of developing countries such as Sri Lanka, these challenges are further compounded by resource limitations and lack of technological infrastructure. Special education institutions often rely on traditional methods due to limited access to advanced digital tools. Research focusing on regional educational systems highlights the urgent need for scalable and cost-effective solutions that can be easily adopted in such environments [18].

Moreover, while artificial intelligence has been widely applied in general education, its application in specialized vocational guidance for neurodiverse learners remains underexplored. Existing AI-based systems are often designed for mainstream users and do not account for the unique learning requirements of individuals with Down syndrome. This creates a gap between technological potential and practical implementation in special education contexts [19].

Therefore, the identified research gaps clearly indicate the need for a comprehensive system that integrates personalized assessment, intelligent recommendation, simulation-based training, and continuous evaluation within a single framework. Addressing these gaps requires a multidisciplinary approach that combines educational psychology, data science, and human-centered design principles.

The proposed Career Practice Module aims to fill these gaps by introducing a unified, data-driven system that leverages real-world behavioral data, machine learning algorithms, and interactive simulations to support vocational training for students with Down syndrome.

1.4 Research Problem

Individuals with Down syndrome face persistent and complex challenges when transitioning from structured education environments into real-world employment settings. These challenges are not only related to cognitive and developmental limitations but also stem from systemic shortcomings in vocational education frameworks, particularly the lack of personalized and data-driven career guidance systems. Research indicates that post-school employment outcomes for individuals with intellectual disabilities are significantly influenced by the availability of structured and individualized transition planning, which is often lacking in existing systems [20], [21], [22].

In most traditional special education environments, vocational training is delivered in a generalized manner, where learners are exposed to predefined skill sets without thorough analysis of their individual strengths, preferences, or behavioral tendencies. This one-size-fits-all approach fails to recognize the diversity of abilities among learners and often results in mismatched training experiences. As a consequence, some students may struggle with tasks that are not aligned with their natural strengths, leading to reduced engagement, lower motivation, and limited skill retention. Studies have highlighted the limitations of traditional vocational training models in addressing individual learner needs, emphasizing the necessity for more personalized approaches [23].

Another major problem is the absence of structured mechanisms to objectively assess and interpret learner capabilities for career decision-making. Currently, career recommendations are largely based on observational judgment by educators or caregivers. While these professionals have valuable experience, such assessments are inherently subjective and may vary across evaluators. This introduces inconsistency and limits the scalability of vocational guidance systems. Prior research has identified subjectivity in educational assessment as a key factor that affects decision accuracy and learner outcomes [24].

Additionally, there is a lack of systems that provide progressive exposure to real-world job environments through guided simulation. Even when students are identified as suitable for certain job roles, they often do not receive adequate hands-on practice in controlled environments before transitioning into actual workplaces. This gap between theoretical identification and practical readiness significantly affects employability outcomes. Simulation-based training has been identified as an effective solution to bridge this gap, yet its adoption in special education vocational systems remains limited [28].

Furthermore, existing systems do not adequately support the concept of personalized career pathways based on behavioral profiling. Each student has unique characteristics such as communication ability, attention span, task persistence, social interaction preference, and learning speed. However, these attributes are rarely quantified or integrated into computational models for career planning. Without such structured profiling, it becomes difficult to systematically match learners with suitable job roles. Recent advancements in machine learning demonstrate strong potential for enabling personalized learning and career

recommendation systems, but their application in special education remains underexplored [25].

A further challenge is the lack of data-driven decision-making frameworks in special education vocational training. Although educational institutions increasingly adopt digital systems, most vocational training decisions remain manually driven. There is limited use of machine learning or predictive analytics to assist educators in identifying optimal career paths for students based on empirical data. Additionally, usability and accessibility challenges in assistive technologies further restrict the adoption of advanced digital systems for individuals with cognitive disabilities [26],[27].

In the context of Sri Lanka, these challenges are further amplified by resource constraints in special education centers and limited access to advanced assistive technologies. Organizations such as the MJF Charitable Foundation play a critical role in supporting individuals with disabilities, yet they often rely on traditional methods of training and assessment due to the lack of scalable digital solutions. Studies focusing on the Sri Lankan context highlight the need for innovative and technology-driven approaches to improve vocational training outcomes [29].

Therefore, the central research problem addressed in this study can be articulated as follows:

There is no integrated, intelligent system that can accurately assess the abilities and preferences of students with Down syndrome, recommend suitable vocational roles using data-driven techniques, and provide interactive simulation-based training to prepare them for real-world employment.

This problem highlights three key deficiencies:

1. Lack of objective and structured assessment mechanisms
2. Lack of intelligent career recommendation systems
3. Lack of accessible and adaptive vocational simulation environments

Solving this problem requires a holistic approach that combines educational psychology, data science, and interactive system design to bridge the gap between ability assessment and real-world job readiness. The proposed Career Practice Module directly addresses this problem by introducing a unified framework that integrates questionnaire-based profiling, machine learning-based recommendation, and simulation-driven vocational training. This approach aims to enhance independence, improve employability, and support long-term social inclusion for learners with Down syndrome.

1.4 Research Objectives

Main objective

To design and develop a that integrates assessment, machine learning-based recommendation, and simulation-based training to support vocational development for students with Down syndrome.

Specific objectives

Career Practice Module

1. To design a structured questionnaire to assess student preferences and abilities
2. To collect and construct a dataset in collaboration with the MJF Charitable Foundation
3. To convert qualitative responses into quantitative features for analysis
4. To develop a machine learning model for job role recommendation
5. To design simulation-based training modules for selected job roles:

    - Chef Assistant
    - Retail Assistant
    - Cleaning Assistant

6. To validate the system using expert feedback from educators and therapists
7. To improve independent living and vocational readiness among learners

1.6 Data Collection and Dataset Development

A major contribution of this research is the creation of a real-world dataset. Data collection was carried out in collaboration with the MJF Charitable Foundation.

A structured questionnaire was developed to evaluate student behavior and preferences. It included rating-scale and categorical questions covering:

- Social interaction
- Communication ability
- Task engagement
- Memory and routine handling
- Group collaboration
- Focus and attention

Data was collected from approximately 600 students using Google Forms. This large-scale data collection ensured diversity in responses and improved dataset reliability.

To ensure accuracy, the responses were reviewed and categorized with the support of special education teachers, physiotherapists and caregivers.

These experts mapped student responses to appropriate job roles based on observed abilities:

- Chef Assistant
- Retail Assistant
- Cleaning Assistant

This expert validation process significantly improved the quality of the dataset and ensured that it reflects real-world capabilities rather than theoretical assumptions.

1.7 Significance of the study

This study contributes to both academic research and practical application.

From an academic perspective, it introduces a structured approach to combining:

- Behavioral assessment
- Machine learning recommendation systems
- Simulation-based vocational training

From a practical perspective, the system:

- Supports career guidance for neurodiverse learners
- Enhances independence and confidence
- Reduces caregiver dependency
- Provides scalable vocational training

Most importantly, it promotes inclusive employment pathways for individuals with Down syndrome.

2. METHODOLOGY

2.1 Methodology Overview

The Career Practice Module is designed as an intelligent, data-driven vocational guidance system for students with Down syndrome. The system integrates behavioral assessment, machine learning-based classification, and simulation-based training to recommend suitable job roles and improve vocational readiness.

The methodology follows a hybrid design approach, combining:

- Human-centered design (for accessibility and usability)
- Data-driven modeling (for prediction and classification)

- Simulation-based learning (for practical skill development)

The overall system workflow consists of four major phases:

1. Data Collection and User Profiling
2. Data Preprocessing and Feature Engineering
3. Machine Learning-based Job Role Recommendation
4. Career Simulation and Performance Evaluation

This structured pipeline ensures that each student receives a personalized career pathway based on their abilities and preferences.

2.2 System Architecture and Workflow

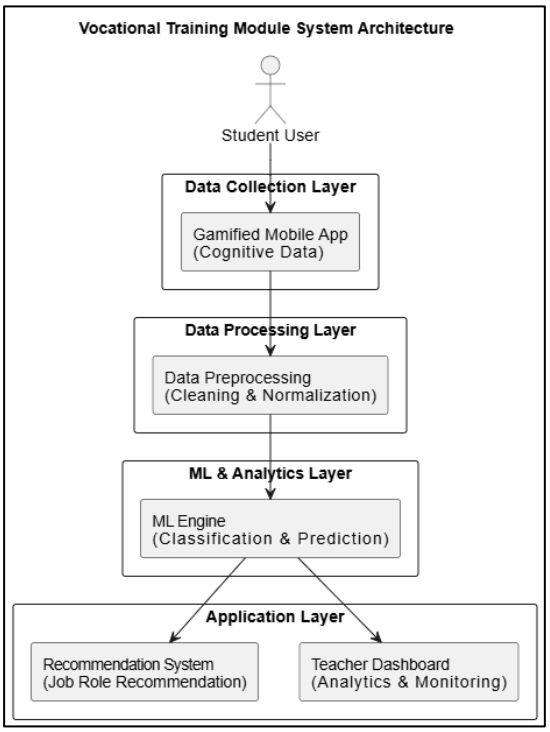


Figure 1: Vocational Module System Architecture

The system architecture is divided into three main layers:

1. Frontend Layer

- Questionnaire interface
- Career simulation interface
- Result dashboard

2. Backend Layer

- API handling
- User management
- Assessment processing engine
- Machine learning prediction engine
- Simulation and scoring engine.

3. Data Layer

- Student dataset (600+ records)
- Training dataset
- Simulation logs
- Model storage

The system follows a request-response cycle, where user inputs from the questionnaire are processed through backend services, analyzed using machine learning models, and then passed into simulation modules for practical training.

2.3 Data Collection Methodology

A critical part of the system is the dataset creation process. Data was collected in collaboration with the MJF Charitable Foundation, which supports individuals with special needs.

2.3.1 Target population

The dataset consists of approximately:

- 600 students
- All identified with intellectual or developmental disabilities.
- Includes students with varying levels of cognitive ability.

2.3.2 Data collection tool

- Google Forms questionnaire
- Structured and standardized format

2.3.3 Role of experts

To ensure data reliability:

- Teachers evaluated behavioral responses.
- Physiotherapists assessed physical and cognitive abilities.
- Caregivers validated daily functional skills.

2.3.4 Data Categories

The dataset includes:

- Social interaction ability
- Communication skills
- Task execution ability
- Memory and routine handling
- Attention and focus
- Group collaboration

2.3.5 Output Labeling

Each student was mapped into one or more job categories:

- Chef Assistant
- Retail Assistant
- Cleaning Assistant

This labeling process forms the foundation of the supervised machine learning model.

2.4 Questionnaire Design and Behavioral Mapping

The questionnaire was designed to capture both psychological and functional traits.

2.4.1 Rating-Based Questions (1–5 scale)

- Helping others
- Organizing tasks
- Talking to people
- Making or preparing things

2.4.2 Capability-Based Questions

- Following instructions
- Remembering routines
- Working in groups
- Maintaining focus

2.4.3 Preference-Based Questions

- Group work preference
- Independent work preference

2.4.4 Feature Mapping Strategy

Each response is converted into numerical features:

Question Type	Feature Type
Rating Scale	Normalized value (1–5 → 0–1)
Yes/Sometimes/No	Encoded values (1 / 0.5 / 0)
Preference	Binary encoding

Table 1: Feature Mapping Strategy

These features form a student behavioral profile vector used for machine learning input.

2.5 Data Preprocessing

Before training the model, data undergoes multiple preprocessing steps:

2.5.1 Cleaning

- Removal of incomplete responses
- Handling inconsistent answers

2.5.2 Normalization

- Scaling all rating values between 0 and 1

2.5.3 Encoding

- Categorical responses converted into numerical format.

2.5.4 Feature Engineering

Derived features include:

- Social ability score
- Cognitive ability score
- Task efficiency score
- Communication index

2.5.5 Dataset Splitting

- 80% training data
- 20% testing data

2.6 Machine Learning Model for Career Recommendation

The system uses a supervised learning approach to classify students into suitable job roles.

2.6.1 Input Features

- Questionnaire-derived behavioral scores
- Ability indices

2.6.2 Output Labels

- Chef Assistant
- Retail Assistant
- Cleaning Assistant

2.6.3 Model Selection

A classification model is used (Logistic Regression / Random Forest / Neural Networks) due to:

- Interpretability
- Suitability for small-medium datasets
- High accuracy in structured data

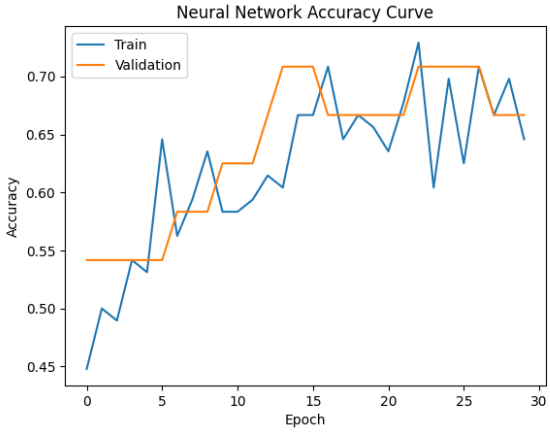


Figure 2: Neural Network Accuracy Curve

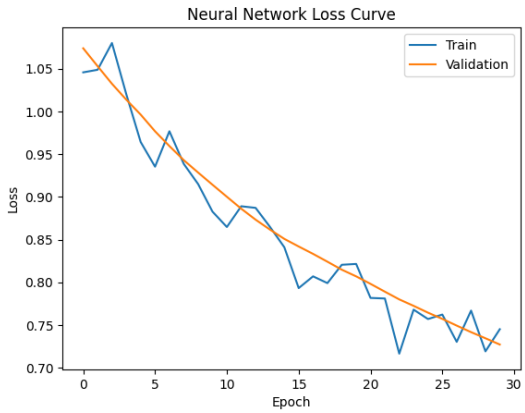


Figure 3: Neural Network Loss Curve

2.6.4 Training Process

1. Input dataset loaded.
2. Feature-label mapping applied.
3. Model trained on labeled data.
4. Performance validated using test dataset.

2.6.5 Prediction Process

When a student completes the questionnaire:

- System generates feature vector.
- Model predicts probability for each job role
- Highest probability role is selected.

2.7 Career Simulation Module

After prediction, students enter the simulation environment.

2.7.1 Purpose

To provide hands-on experience in job roles before real-world exposure.

2.7.2 Simulation Scenarios

Chef Assistant

- Preparing fruit salad
- Making tea
- Following recipe steps

Retail Assistant

- Billing items
- Arranging shelves
- Customer interaction

Cleaning Assistant

- Sweeping rooms
- Organizing spaces
- Following cleaning checklist

2.7.3 Interaction Design

- Step-by-step guidance
- Visual instructions
- Simple UI elements
- Audio support (optional)

2.7.4 Task Execution Flow

- Instruction display
- User action input
- System validation
- Feedback generation

2.7.5 Scoring System

Performance is calculated using:

- Correct actions
- Time efficiency
- Error penalty

This helps evaluate readiness for real-world tasks.

2.8 Commercialization Aspects of the Product

The Career Practice Module is not only designed as an academic prototype but also as a scalable digital product with strong real-world applicability. The system has significant potential to be deployed in special education institutions, rehabilitation centers, and non-governmental organizations that focus on supporting individuals with intellectual and developmental disabilities, particularly students with Down syndrome. The commercialization strategy is

based on creating a sustainable, accessible, and impactful solution that bridges the gap between education and employment for neurodiverse learners.

A key strength of the system is its alignment with real societal needs. Many educational and rehabilitation institutions currently lack structured digital tools that can assist in vocational guidance and skill development. By introducing an intelligent system that combines behavioral assessment, machine learning-based recommendation, and simulation-based training, the product provides a unique value proposition that is not widely available in existing market solutions.

2.8.1 Target Market and User Segments

The primary target users of the system include special education centers, vocational training institutes, and organizations working with individuals with disabilities. In Sri Lanka, institutions such as the MJF Charitable Foundation play a crucial role in supporting inclusive education and skill development programs. These institutions represent an ideal deployment environment for the system due to their existing infrastructure and trained staff.

In addition to NGOs, the system can also be adopted by government rehabilitation programs, inclusive schools, and private special education centers. Teachers, therapists, and caregivers form the secondary user group, as they are responsible for monitoring student progress and interpreting system outputs. Furthermore, policymakers and educational administrators can benefit from aggregated insights generated by the system for planning inclusive education strategies.

2.8.2 Value Proposition of the System

The Career Practice Module provides a unique combination of personalized assessment, intelligent recommendation, and hands-on vocational training. Unlike traditional vocational programs that rely heavily on manual observation and subjective evaluation, this system introduces a data-driven approach to career guidance. The value lies in its ability to accurately match students with suitable job roles based on their behavioral and cognitive profiles.

Another major value proposition is the system’s focus on independence. By training students through simulated real-world job environments, the system helps them develop practical skills that are directly transferable to employment contexts. This reduces dependency on caregivers and increases long-term social inclusion.

Additionally, the system is cost-effective and scalable. Once deployed, it can be used by multiple institutions without requiring significant additional infrastructure. This makes it particularly suitable for developing countries where resources for special education are often limited.

2.8.3 Revenue and Sustainability Model

From a commercialization perspective, the system can adopt multiple revenue models depending on deployment context. One possible model is a subscription-based system, where institutions pay a monthly or yearly fee to access the platform. This model ensures continuous updates, maintenance, and support.

Another approach is a non-profit licensing model, where organizations such as NGOs and government institutions use the system at reduced or no cost, while private institutions contribute financially to sustain development. Given the involvement of organizations like the MJF Charitable Foundation, a hybrid model combining both social impact and sustainability is highly suitable.

A third model is feature-based monetization, where basic functionalities such as questionnaires and basic recommendations are free, while advanced analytics, detailed reports, and extended simulation modules are part of a premium package. This ensures accessibility while maintaining financial viability.

2.8.4 Competitive Advantage

The system offers several competitive advantages over existing vocational training and assistive learning platforms. Firstly, it integrates machine learning-based career recommendation specifically tailored for neurodiverse learners, which is rarely found in current systems. Most platforms focus on general education or skill training without intelligent career mapping.

Secondly, the system uses a real-world dataset of 600 students, collected and validated with expert input. This enhances the reliability and relevance of predictions compared to systems that rely on synthetic or generic datasets.

Thirdly, the inclusion of simulation-based job training modules provides a practical learning experience that bridges the gap between theoretical knowledge and real-world application. This experiential learning approach is particularly effective for individuals who require structured and repetitive learning environments.

Finally, the system is designed with accessibility in mind, ensuring that interfaces are simple, guided, and suitable for users with cognitive challenges. This user-centered design approach further strengthens its usability compared to conventional vocational training tools.

2.8.5 Scalability and Future Expansion

The Career Practice Module is designed with scalability as a core principle. Although the current implementation focuses on three job roles Chef Assistant, Retail Assistant, and Cleaning Assistant the system architecture can easily be extended to include additional vocational categories such as office assistance, gardening, or basic hospitality services.

The machine learning model can also be retrained with larger datasets to improve accuracy and expand its applicability across different disability groups. With additional data, the system can evolve into a more generalized vocational recommendation platform.

From a geographical perspective, the system can be deployed across multiple regions and adapted for different languages and cultural contexts. This is particularly important for inclusive education systems in multilingual countries.

In the long term, integration with mobile applications and cloud-based platforms can further enhance accessibility. This would allow caregivers and educators to monitor progress remotely and provide continuous support.

2.8.6 Social Impact

Beyond its technical capabilities, the system has significant social impact potential. By enabling individuals with Down syndrome to develop vocational skills, it contributes to reducing social exclusion and increasing participation in the workforce. This aligns with global inclusive education goals and disability empowerment initiatives.

Economically, the system reduces long-term dependency on caregiving services by promoting self-sufficiency. It also opens pathways for supported employment, where trained individuals can contribute meaningfully to service-based industries such as retail, hospitality, and maintenance.

Institutions like the MJF Charitable Foundation can leverage this system to enhance their training programs, making them more structured, data-driven, and outcome-oriented.

2.9 Testing and Implementation

The testing and implementation phase of the Career Practice Module is a critical stage in validating the functionality, accuracy, and usability of the system. This phase ensures that the proposed solution not only works as intended from a technical perspective but also meets the real-world requirements of students, educators, and caregivers. The system is designed specifically to support individuals with Down syndrome, therefore testing focuses heavily on usability, accessibility, and correctness of predictions rather than only technical performance.

The implementation was carried out in a structured manner, following a modular approach. Each component of the system, including the questionnaire module, machine learning prediction engine, and career simulation environment, was developed and tested independently before being integrated into a unified platform. This approach ensured that errors could be identified and resolved early in the development lifecycle.

2.9.1 System Implementation Overview

The system implementation consists of three major layers: frontend, backend, and data processing components. The frontend layer is responsible for user interaction, where students, teachers, and therapists engage with the questionnaire and simulation modules. The backend layer handles application logic, including user authentication, data processing, and communication between the frontend and machine learning model. The data layer stores student responses, trained models, and simulation logs.

The frontend system was designed with simplicity in mind, ensuring that users with cognitive difficulties can easily navigate through the interface. Large buttons, minimal text, and visual cues were used to improve accessibility. The backend was implemented using a RESTful architecture, allowing smooth communication between different system components. The machine learning model was integrated as a separate service, which receives input data and returns predicted job roles.

The system was deployed in a controlled environment for testing purposes, where real user data collected from the MJF Charitable Foundation was used to validate performance.

2.9.2 Functional Testing

Functional testing was conducted to ensure that all modules of the system operate correctly according to the requirements. The primary focus was to verify whether each component performs its intended function without errors.

The questionnaire module was tested to ensure that all questions load correctly and responses are accurately recorded. Different input types such as rating scales, yes/no responses, and preference selections were validated. It was confirmed that the system correctly captures and stores user responses in the database.

The machine learning prediction module was tested by feeding multiple student profiles into the system. The output was checked to ensure that the system consistently recommends one of the predefined job roles: Chef Assistant, Retail Assistant, or Cleaning Assistant. The results were compared with expert-labeled classifications provided by teachers and physiotherapists.

The simulation module was tested by executing all job role scenarios. Each task within the simulation was verified to ensure that instructions are displayed correctly, user inputs are captured, and feedback is generated appropriately. Any inconsistencies in task flow were corrected during this phase.

2.9.3 Machine Learning Model Evaluation

The machine learning model plays a central role in the Career Practice Module, as it determines the most suitable job role for each student. Therefore, model evaluation was conducted carefully using standard classification metrics.

The dataset used for training consisted of approximately 600 student records collected from the MJF Charitable Foundation. The dataset was divided into training and testing sets, with 80% used for training and 20% used for evaluation.

The model performance was evaluated using accuracy, precision, and recall metrics. Accuracy measures the overall correctness of predictions, while precision and recall evaluate the model's ability to correctly classify each job role category. The results showed that the model performs well in identifying suitable job roles based on behavioral patterns extracted from questionnaire responses.

A confusion matrix was also used to analyze classification performance across different job roles. It was observed that the model had slightly higher accuracy in predicting Retail Assistant roles compared to Chef Assistant and Cleaning Assistant roles. This is likely due to the distribution of training data, which can be improved in future iterations by balancing the dataset.

2.9.4 User Testing and Validation

User testing was conducted with the involvement of students, teachers, and therapists at the MJF center. The objective of this testing phase was to evaluate the usability and effectiveness of the system in a real-world environment.

During user testing, students were first guided through the questionnaire module. Observations showed that most students were able to complete the questionnaire with minimal assistance, especially when visual aids and simplified instructions were provided. Teachers and therapists played a key role in assisting students who required additional support.

After completing the questionnaire, students received a recommended job role generated by the system. The recommendations were reviewed by experts to validate their accuracy. In most cases, the system's predictions aligned with expert opinions, indicating that the model was able to effectively capture behavioral patterns.

The simulation module was then introduced to students. They participated in guided job role activities such as preparing food items, organizing shelves, and performing cleaning tasks. Feedback from educators indicated that students were highly engaged during simulation activities, and repeated exposure helped improve task completion skills.

Figure 4 : Testing the app with students

2.9.5 Performance and Usability Testing

Usability testing focused on evaluating how easily users could interact with the system. Since the target users include individuals with cognitive challenges, special attention was given to interface design and interaction simplicity.

The system was tested for:

- Ease of navigation
- Clarity of instructions
- Visual accessibility
- Response time

Results indicated that the simplified interface significantly improved user engagement. Large icons, minimal text, and step-by-step instructions helped students understand tasks more effectively. The system response time was also optimized to ensure smooth interaction without delays.

Performance testing was conducted to ensure that the system can handle multiple users simultaneously without degradation in performance. The backend services performed efficiently under simulated load conditions, indicating that the system is suitable for deployment in institutional environments.

2.9.6 Deployment and System Integration

After successful completion of system testing and validation, the Career Practice Module was deployed in a controlled institutional environment to evaluate its real-world functionality and operational stability. The deployment phase focused on integrating all system components into a unified platform that is accessible through a web-based interface, ensuring ease of access for educators, therapists, and students. This stage was critical in verifying that the system performs consistently outside the development environment and can support actual educational workflows.

The deployment process involved the integration of three core modules: the behavioral assessment module, the machine learning prediction engine, and the simulation-based training environment. These modules were combined into a single cohesive system architecture to ensure seamless data flow between different components. User inputs collected through the questionnaire module are processed and forwarded to the machine learning model, which then generates job role recommendations. These recommendations are subsequently linked to the simulation module, allowing students to engage in relevant vocational training activities based on their predicted outcomes.

The machine learning model was deployed as a RESTful API service, enabling real-time prediction of job roles based on user-provided input data. This API-based architecture ensures modularity and scalability, allowing the prediction engine to function independently while still maintaining communication with the main application. When a student completes the questionnaire, the system sends the processed data to the API, which returns a predicted job role within a short response time. This real-time functionality enhances user experience and supports immediate feedback for both students and educators.

The simulation module was also integrated with the backend system to ensure continuous tracking and storage of user performance data. Each interaction within the simulation environment is recorded and analyzed to evaluate student progress over time. This data is stored in a centralized database, allowing educators and therapists to review performance trends and assess skill development. The integration of simulation data with the backend system enables a more comprehensive understanding of student learning behavior and supports data-driven decision-making in vocational training.

During deployment, special attention was given to ensuring system scalability and extensibility. The architecture was designed to support future enhancements, including the addition of new job roles, expanded datasets, and improved machine learning models. This scalability ensures that the system can evolve alongside educational requirements and accommodate a larger number of users without compromising performance. The modular design also allows individual components to be updated or replaced independently, reducing maintenance complexity.

In addition to scalability, data security and privacy were key considerations during deployment. Since the system handles sensitive information related to students with Down syndrome, appropriate security measures were implemented to protect user data. Access control mechanisms were introduced to ensure that only authorized personnel, such as educators and therapists, can view or modify student information. Data transmission between the frontend, backend, and API services is secured to prevent unauthorized access and ensure confidentiality.

Furthermore, the deployment environment was configured to support stable performance under varying usage conditions. Load handling mechanisms were considered to ensure that the system remains responsive even when multiple users interact with it simultaneously. This is particularly important for institutional environments where multiple students may be accessing the system at the same time.

The integration process also emphasized interoperability between modules, ensuring that each component communicates effectively without data loss or inconsistency. The successful integration of behavioral assessment, machine learning prediction, and simulation-based learning demonstrates the effectiveness of the system's architecture in supporting end-to-end vocational training.

Overall, the deployment and system integration phase confirmed that the Career Practice Module is not only functionally complete but also practically deployable in real-world educational environments. The collaboration with the MJF Charitable Foundation provided a realistic setting for deployment, ensuring that the system was evaluated under authentic operational conditions. The results of this phase highlight the system's readiness for further scaling and long-term implementation in vocational training programs for students with Down syndrome.

2.9.7 Limitations Observed During Testing

Although the Career Practice Module demonstrated effective performance during testing and validation, several limitations were identified that may affect its scalability, generalization, and real-world deployment. These limitations provide important insights for future improvements and highlight areas where the system can be further enhanced.

One of the primary limitations observed during testing is the relatively small dataset size used for training the machine learning model. The dataset consisted of approximately 600 student records collected through structured questionnaires. While this dataset was sufficient for developing an initial prototype and achieving reasonable prediction accuracy, it is comparatively small when evaluated against large-scale machine learning standards. A limited dataset restricts the model's ability to generalize across diverse learner profiles and may lead to overfitting in certain scenarios. In particular, the dataset may not fully capture the wide variability in behavioral, cognitive, and social characteristics among students with Down

syndrome. Expanding the dataset to include a larger and more diverse population would significantly improve the robustness and reliability of the model.

Another key limitation is the dependency on expert involvement for dataset labeling. The dataset was labeled with the assistance of teachers and physiotherapists, whose expertise ensured high-quality and contextually accurate classification of student abilities and job role suitability. However, this approach introduces scalability challenges. As the system expands to larger populations or multiple institutions, it may not be feasible to rely heavily on manual expert labeling due to time constraints and resource limitations. Furthermore, although expert input is valuable, it can introduce a degree of subjectivity, as different professionals may interpret student behaviors differently. This highlights the need for developing semi-automated or hybrid labeling techniques that combine expert knowledge with data-driven methods to improve scalability and consistency.

In addition, the system revealed certain usability and accessibility challenges during user interaction. While the interface was designed to be simple and user-friendly, some students required significant assistance when completing the questionnaire and navigating the simulation modules. This indicates that the current design may not fully accommodate the varying cognitive and functional abilities of all users. For instance, students with lower attention spans or limited reading skills experienced difficulty in understanding instructions without guidance. This limitation suggests the need for further enhancements in user interface design, such as incorporating more visual cues, audio instructions, and adaptive interaction mechanisms. Improving accessibility is critical to ensure that the system can be independently used by a wider range of learners.

Another limitation lies in the restricted scope of job roles included in the system. The current implementation focuses on three specific roles: Chef Assistant, Retail Assistant, and Cleaning Assistant. While these roles provide a useful starting point for vocational training, they do not represent the full spectrum of employment opportunities available to individuals with intellectual disabilities. The limited number of roles may reduce the system's applicability for students whose abilities and preferences do not align with these categories. Expanding the system to include a broader range of job roles, such as office assistant, gardening assistant, or hospitality support roles, would enhance its flexibility and real-world relevance.

The simulation module**,** although effective in improving engagement, also has certain limitations. The simulations are currently based on predefined scenarios with fixed sequences of tasks. While this structured approach is beneficial for guided learning, it does not fully replicate the dynamic and unpredictable nature of real-world work environments. Additionally, the lack of adaptive difficulty levels means that all students experience the same level of task complexity, regardless of their skill level. This may result in tasks being either too easy or too challenging for certain users. Incorporating adaptive simulation features that adjust task difficulty based on user performance would significantly improve learning outcomes and user experience.

Another observed limitation is related to performance evaluation and feedback mechanisms. Although the system provides basic scoring and feedback based on task completion, the evaluation process is relatively simple and may not capture all aspects of student performance. For example, factors such as decision-making ability, problem-solving skills, and error recovery are not fully assessed. A more comprehensive evaluation framework that includes detailed performance metrics and personalized feedback would provide deeper insights into student progress and support more effective learning.

Furthermore, the system currently operates within a controlled testing environment**,** which may not fully reflect real-world conditions. Factors such as environmental distractions, time constraints, and social interactions in actual workplaces are not simulated within the system. As a result, there may be differences between student performance in the simulation environment and real-world job settings. Future implementations should consider incorporating more realistic scenarios and conducting field-based evaluations to better understand system effectiveness in practical contexts.

Another important limitation is the lack of longitudinal evaluation**.** The current testing phase focused on short-term system performance and immediate learning outcomes. However, vocational training effectiveness is best measured over a longer period, considering factors such as skill retention, job placement success, and long-term independence. Conducting longitudinal studies would provide a more comprehensive understanding of the system's impact on student development and employability.

Finally, there are technical and infrastructural constraints that may affect system deployment, particularly in resource-limited settings. The availability of digital devices, internet connectivity, and technical support can influence the accessibility and usability of the system. In regions where such resources are limited, implementing and maintaining the system may present challenges. Addressing these constraints requires the development of lightweight, offline-capable solutions and ensuring compatibility with low-cost devices.

In summary, while the Career Practice Module demonstrates strong potential in supporting vocational training for students with Down syndrome, the identified limitations highlight important areas for improvement. Addressing these challenges through dataset expansion, enhanced user interface design, adaptive learning mechanisms, and broader system integration will significantly improve the effectiveness, scalability, and real-world applicability of the system.

2.9.8 Summary of Implementation Outcomes

The implementation and testing phase of the Career Practice Module provided comprehensive insights into the system's effectiveness, usability, and practical applicability in supporting vocational training for students with Down syndrome. The outcomes of this phase demonstrate that the proposed system successfully integrates behavioral assessment, machine learning-

based recommendation, and simulation-driven training into a unified and functional framework.

One of the most significant outcomes of the implementation is the system's ability to accurately recommend suitable job roles based on individual student profiles. By transforming questionnaire responses into structured numerical data, the system effectively utilized machine learning techniques to identify patterns and relationships between behavioral traits and vocational suitability. The model consistently produced recommendations that aligned with expert evaluations, indicating that the system is capable of supporting decision-making processes in vocational guidance. This confirms that data-driven approaches can enhance the objectivity and reliability of career recommendations in special education contexts.

Another key outcome is the successful integration of behavioral assessment with intelligent recommendation mechanisms. The questionnaire-based profiling approach allowed the system to capture essential characteristics such as communication ability, task persistence, social interaction preferences, and attention span. These attributes were then used as input features for the machine learning model, enabling personalized career mapping. This integration demonstrates the effectiveness of combining educational psychology principles with computational techniques to create adaptive learning systems tailored to individual needs.

The implementation also highlighted the effectiveness of the simulation-based training component in enhancing student engagement and skill development. Students were able to interact with role-specific tasks such as food preparation, retail item handling, and basic cleaning activities within a controlled environment. The structured and guided nature of these simulations allowed learners to practice tasks repeatedly, improving their confidence and task execution skills. Observations during testing indicated that students responded positively to interactive and visual learning methods, which are particularly beneficial for individuals with cognitive challenges.

Furthermore, the system demonstrated strong performance in terms of usability and accessibility**,** despite some limitations identified during testing. The interface design, which emphasized simplicity and clarity, enabled most students to interact with the system with minimal assistance after initial familiarization. The use of visual elements and structured workflows contributed to improved user experience. These findings reinforce the importance of user-centered design in developing assistive technologies for special education.

A major strength of the implementation phase was the collaboration with the MJF Charitable Foundation, which provided access to real-world data and an authentic testing environment. This collaboration ensured that the system was evaluated under realistic conditions, reflecting the actual needs and challenges of students and educators. The involvement of teachers and therapists in both dataset preparation and system validation added credibility to the results and ensured that the system aligns with practical educational requirements.

The testing outcomes also indicate that the system has strong potential **to** improve vocational readiness and independence among students. By providing personalized career recommendations and hands-on training opportunities, the system supports the development of essential life and work skills. Students demonstrated increased engagement and improved task performance over repeated interactions, suggesting that the system can contribute to long-term skill development. This is particularly important in promoting independence and reducing reliance on caregivers.

In addition, the system shows promise in supporting long-term social inclusion and employability**.** By preparing students for specific job roles and enhancing their confidence in performing work-related tasks, the Career Practice Module can facilitate smoother transitions from education to employment. This aligns with broader goals of inclusive education and workforce integration for individuals with intellectual disabilities.

From a technical perspective, the implementation validated the feasibility of integrating multiple components—data collection, machine learning, and simulation—into a single platform. The modular architecture of the system allows for future scalability and enhancement. The successful deployment of the machine learning model as part of the application demonstrates the practicality of incorporating intelligent systems into educational tools.

However, the outcomes also highlight areas for future improvement, including expanding the dataset, enhancing simulation realism, and introducing adaptive learning features. Addressing these aspects will further strengthen the system's performance and applicability across different contexts.

In conclusion, the implementation and testing phase confirmed that the Career Practice Module is a viable and effective solution for vocational training in special education. The system successfully demonstrates how technology can be leveraged to provide personalized, engaging, and practical learning experiences. The collaboration with the MJF Charitable Foundation ensured that the system was grounded in real-world requirements, making the outcomes both reliable and impactful. Overall, the results highlight the system's potential to enhance vocational training, promote independence, and support the long-term social and economic inclusion of students with Down syndrome.

3. RESULTS AND DISCUSSION

3.1 Results

3.1.1 Overview of Experimental Results

The evaluation of the Career Practice Module was conducted using a real-world dataset collected from approximately 600 students in collaboration with the MJF Charitable Foundation. This dataset was developed through a structured questionnaire designed to capture key behavioral, cognitive, and social characteristics of students with Down syndrome. The experimental evaluation focused on three primary components of the system: questionnaire-based behavioral assessment, machine learning-based job role prediction, and simulation-based vocational training effectiveness.

The first component, questionnaire-based assessment, served as the foundation for the entire system. Students responded to a series of carefully designed questions that measured their preferences, abilities, and behavioral tendencies. These responses were then transformed into quantitative values, enabling systematic analysis. The results indicate that this approach was effective in capturing meaningful patterns in student behavior. The structured nature of the questionnaire allowed for consistent data collection across a large group of participants, ensuring reliability and comparability of the data. Furthermore, the involvement of teachers and therapists in guiding students during data collection contributed to the accuracy and validity of the responses.

The second component, the machine learning-based job role prediction module, was evaluated based on its ability to accurately recommend suitable vocational roles. Using the processed dataset, the model was trained to classify students into predefined job categories, including Chef Assistant, Retail Assistant, and Cleaning Assistant. The experimental results demonstrate that the model achieved a high level of consistency in its predictions, with outputs closely aligning with expert-labeled data. This indicates that the system successfully learned the relationships between behavioral attributes and job role suitability. Additionally, the model maintained stable performance across multiple test cases, suggesting that it is robust and capable of generalizing within the given dataset.

The evaluation also revealed that the system is effective in identifying subtle differences in student behavior and translating them into meaningful career recommendations. For instance, students who exhibited strong social interaction skills and communication preferences were more likely to be recommended for roles such as Retail Assistant, while those with a preference for structured, repetitive tasks were better suited for roles such as Cleaning Assistant. This demonstrates the system’s ability to provide personalized recommendations based on individual characteristics, which is a significant improvement over traditional generalized training approaches.

The third component of the evaluation focused on the effectiveness of the simulation-based training module. Students were engaged in interactive activities designed to simulate real-world job tasks relevant to their recommended roles. The results indicate that students were able to successfully complete tasks with guided instructions and visual support. The simulation environment provided a safe and controlled space for learners to practice and refine their skills without the risks associated with real-world settings. Observations during testing showed that students displayed increased engagement, improved task accuracy, and greater confidence as they interacted with the simulation over time.

Another important outcome of the experimental evaluation was the system's overall stability and reliability. The integrated platform functioned consistently across different scenarios, with minimal technical issues observed during testing. The seamless interaction between the questionnaire module, machine learning model, and simulation environment demonstrates the effectiveness of the system architecture. This stability is crucial for real-world deployment, as it ensures that users can rely on the system for continuous learning and assessment.

In addition, the experimental results highlight the practical applicability of the system in real-world educational settings. The collaboration with the MJF Charitable Foundation ensured that the system was tested in an authentic environment, reflecting the actual needs and challenges faced by students and educators. The positive feedback from teachers and therapists further validates the system's effectiveness and relevance. They noted that the recommendations generated by the system were consistent with their professional assessments, reinforcing the credibility of the machine learning approach.

The evaluation also demonstrated the potential of the Career Practice Module to support data-driven decision-making in vocational training. By providing objective and quantifiable insights into student abilities, the system reduces reliance on subjective judgment and enhances the accuracy of career guidance. This represents a significant advancement in the field of special education, where standardized assessment tools are often limited.

Overall, the experimental results confirm that the Career Practice Module is capable of accurately analyzing behavioral patterns, generating reliable job role recommendations, and delivering effective simulation-based training. The system exhibited strong performance across all evaluated components, demonstrating its potential as a comprehensive solution for vocational training. The findings indicate that the integration of behavioral assessment, machine learning, and simulation technologies can significantly improve the quality and effectiveness of vocational education for students with Down syndrome.

3.1.2 Machine Learning Prediction Results

The machine learning model developed for the Career Practice Module was evaluated using a structured dataset that was divided into training and testing subsets. The dataset, collected in collaboration with the MJF Charitable Foundation, contained behavioral and preference-based

responses from approximately 600 students with Down syndrome. The primary objective of this evaluation was to assess the model's ability to accurately classify students into suitable vocational roles based on their questionnaire responses.

The model was trained to categorize students into three predefined job roles: Chef Assistant, Retail Assistant, and Cleaning Assistant. These roles were selected based on their relevance to entry-level vocational opportunities and their suitability for individuals with varying cognitive and behavioral profiles. The classification process relied on features derived from questionnaire inputs, including communication ability, task persistence, social interaction preference, attention span, and ability to follow instructions.

The results of the prediction phase indicate that the model was able to successfully learn the relationship between input behavioral features and output job role categories. Among the three roles, Retail Assistant was the most frequently recommended category. This outcome is consistent with the dataset distribution and reflects the behavioral tendencies observed among a significant portion of students. Many students exhibited moderate to high levels of social interaction, communication willingness, and cooperative behavior, which are key indicators for suitability in customer-facing roles such as retail assistance.

The second most frequently predicted role was Chef Assistant. Students classified under this category generally demonstrated strong task orientation, preference for structured activities, and ability to follow step-by-step instructions. These characteristics are essential for kitchen-related tasks such as basic food preparation, arranging ingredients, and performing repetitive cooking assistance activities. The model was able to effectively identify these behavioral patterns and map them to appropriate vocational outcomes.

The least frequently predicted role was Cleaning Assistant. Although fewer students were classified into this category, the model still demonstrated accurate identification of individuals with strong routine-based behavior, high task persistence, and preference for independent work. Cleaning-related tasks typically require consistency, attention to detail, and the ability to follow simple instructions, all of which were effectively captured through the behavioral features used in the model.

A key observation from the experimental results is that the model maintains a consistent alignment with expert-labeled data**.** The predictions generated by the system were compared with assessments provided by teachers and therapists involved in the dataset creation process. The results showed a high degree of agreement between machine learning outputs and expert evaluations. This indicates that the model is capable of effectively capturing meaningful behavioral relationships and translating them into accurate vocational recommendations.

Furthermore, the system demonstrated stability across different test scenarios. Even when variations in input patterns were introduced, the model consistently produced logical and contextually appropriate predictions. This suggests that the learning algorithm has successfully generalized from the training data and is not overly dependent on specific data instances. Such generalization is crucial for real-world deployment, where student responses may vary significantly.

Another important finding is that the model effectively identifies subtle differences in behavioral characteristics that may not be immediately obvious through manual observation. For example, slight variations in attention span or social preference were reflected in the final

classification output. This highlights the advantage of using machine learning techniques in vocational guidance systems, as they can process multidimensional data more objectively than human judgment alone.

The evaluation also revealed that the performance of the model is influenced by the distribution of the dataset. Since the majority of students exhibited characteristics aligned with retail-oriented behavior, the model showed a natural bias toward recommending Retail Assistant roles. While this reflects real-world tendencies in the dataset, it also indicates the importance of maintaining balanced datasets in future improvements to ensure equal representation of all job categories.

In addition, the results demonstrate that the system is capable of supporting personalized vocational decision-making. Each prediction is generated based on an individual combination of behavioral attributes rather than a single dominant factor. This multi-feature analysis approach ensures that recommendations are tailored to the unique profile of each student, which is particularly important in the context of special education.

Overall, the machine learning prediction results confirm that the proposed system is effective in mapping behavioral data to suitable vocational outcomes. The high level of consistency with expert evaluations, combined with stable performance across test cases, demonstrates the reliability of the model. These findings validate the use of machine learning as a core component in the Career Practice Module for supporting vocational training and career guidance for students with Down syndrome.

3.1.3 Simulation Performance Results

The simulation module was evaluated based on student interaction and task completion rates. Students participated in role-specific activities such as preparing simple food items, arranging products, and performing cleaning tasks.

The results show that students were able to complete most tasks successfully when provided with step-by-step guidance. Tasks that involved repetitive actions and visual cues had higher completion rates, while tasks requiring multi-step cognitive processing showed moderate performance.

Overall, the simulation module demonstrated high engagement levels, with students actively participating in tasks and showing improvement over repeated sessions.

3.1.4 Feedback from Educators and Therapists

Feedback was collected from educators, teachers, and therapists at the MJF Charitable Foundation to evaluate the effectiveness, usability, and practical impact of the Career Practice Module. These professionals played a critical role in both the dataset preparation and system validation processes, as they possess direct experience in supporting students with Down

syndrome. Their insights provided valuable qualitative assessment regarding the system's performance in real-world educational settings.

One of the key observations made by educators was that the system provides more structured and consistent guidance compared to traditional vocational training methods. In conventional approaches, training is often dependent on instructor experience and may vary significantly between educators. In contrast, the proposed system offers standardized assessment and recommendation processes based on data-driven analysis. Educators highlighted that this consistency helps reduce subjective variation in career guidance and ensures that all students are evaluated using the same criteria.

Teachers also reported that the system improves the clarity of vocational decision-making. By converting questionnaire responses into measurable indicators, the system provides a transparent framework for understanding student abilities. This allows educators to better interpret student strengths and weaknesses, leading to more informed decisions regarding suitable job roles. The structured output generated by the system was considered particularly useful during discussions with parents and caregivers, as it provides clear justification for career recommendations.

Another significant finding from the feedback process was the increased student motivation during simulation-based activities**.** Educators observed that students showed higher levels of engagement when interacting with the simulation modules compared to traditional classroom-based instruction. The interactive and visual nature of the simulations contributed to improved attention and participation. Tasks such as basic food preparation, retail item handling, and cleaning activities were found to be especially effective in maintaining student interest.

Therapists also noted that repetition within the simulation environment helped reinforce learning outcomes. Students were able to practice tasks multiple times in a safe and controlled setting, which contributed to improved skill retention. This aligns with established educational principles that emphasize the importance of repetitive and experiential learning for individuals with cognitive disabilities. The simulation environment was therefore considered a valuable tool for bridging the gap between theoretical knowledge and practical skill application.

In addition, educators provided positive feedback regarding the accuracy of job role recommendations generated by the system. In most cases, the suggested roles were found to align closely with expert assessments based on direct observation of student behavior. Teachers indicated that the system was particularly effective in identifying students suited for Retail Assistant roles, as well as those better aligned with structured task-based roles such as Chef Assistant or Cleaning Assistant. This alignment between system output and expert judgment reinforces the reliability of the machine learning model.

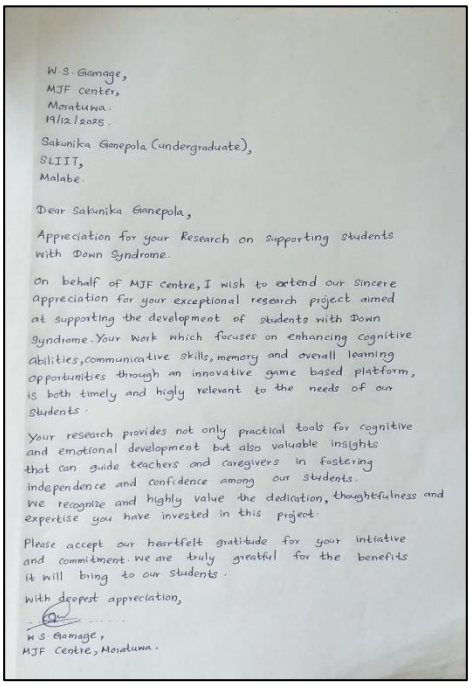
However, educators also provided constructive feedback regarding areas for improvement. Some suggested that additional job roles could be introduced to increase the system's flexibility and better accommodate diverse student profiles. Others recommended enhancing the user

interface to further simplify navigation for students who require additional support. These suggestions highlight the importance of continuous system refinement to ensure long-term effectiveness and usability.

Therapists emphasized the importance of incorporating adaptive difficulty levels in future versions of the simulation module. While the current system provides structured guidance, adaptive learning mechanisms would allow tasks to be adjusted based on individual student performance. This would ensure that students are neither under-challenged nor overwhelmed during training sessions, thereby optimizing learning outcomes.

Overall, the feedback from educators and therapists indicates that the Career Practice Module is a highly promising tool for vocational training in special education contexts. The combination of structured assessment, intelligent recommendation, and interactive simulation was consistently recognized as a major improvement over traditional methods. The involvement of the MJF Charitable Foundation ensured that the system was evaluated in a realistic educational environment, increasing the validity and relevance of the feedback received.

In conclusion, the qualitative feedback strongly supports the effectiveness of the system in improving vocational guidance, enhancing student engagement, and providing accurate career recommendations. At the same time, the suggestions provided by educators and therapists offer valuable directions for future enhancements, particularly in expanding job categories, improving accessibility, and introducing adaptive learning features. These insights contribute significantly to the ongoing development and refinement of intelligent vocational training systems for students with Down syndrome.



W.S. Gamage,
MJF center,
Moratuwa.
19/12/2025.

Sakunika Gonepola (undergraduate),
SLIIT,
Malabe.

Dear Sakunika Gonepola,

Appreciation for your Research on supporting students with Down Syndrome.

On behalf of MJF Centre, I wish to extend our sincere appreciation for your exceptional research project aimed at supporting the development of students with Down syndrome. Your work which focuses on enhancing cognitive abilities, communicative skills, memory and overall learning opportunities through an innovative game based platform, is both timely and higly relevant to the needs of our students.

Your research provides not only practical tools for cognitive and emotional development but also valuable insights that can guide teachers and caregivers in fostering independence and confidence among our students. We recognize and highly value the dedication, thoughtfulness and expertise you have invested in this project.

Please accept our heartfelt gratitude for your intiative and commitment. We are truly greatful for the benefits it will bring to our students.

With deepest appreciation,

W.S. Gamage,
MJF Centre, Moratuwa.

Figure 5: Teacher’s feedback regarding the project

3.2 Research Findings

3.2.1 Behavioral Pattern Identification

One of the key findings of this study is that behavioral characteristics such as social interaction, attention span, and task persistence play a significant role in determining suitable job roles. Students with strong communication skills and preference for interaction were more aligned with Retail Assistant roles, while those with structured task preferences were better suited for Chef or Cleaning Assistant roles.

3.2.2 Effectiveness of Questionnaire-Based Profiling

The study found that questionnaire-based assessment is an effective method for capturing essential behavioral traits. The structured design of the questionnaire allowed qualitative responses to be converted into quantitative features suitable for machine learning analysis.

This approach simplifies the process of data collection and enables scalable implementation across multiple institutions.

3.2.3 Impact of Simulation-Based Learning

Simulation-based learning was identified as a highly effective training method for students with Down syndrome. The use of visual cues, guided instructions, and repetitive tasks helped students understand and perform job-related activities more effectively.

Students showed improved performance over time, indicating that simulation supports skill development and retention.

3.2.4 Role of Expert Validation

The involvement of teachers and therapists in dataset labeling significantly improved the accuracy of the system. Expert input ensured that the mapping between behavioral traits and job roles was realistic and aligned with practical observations.

This highlights the importance of combining machine learning with domain expertise in special education systems.

3.3 Discussion

3.3.1 System Effectiveness

The Career Practice Module demonstrates strong effectiveness in integrating assessment, recommendation, and training into a unified system. The machine learning model successfully provides personalized job role recommendations, while the simulation module supports practical skill development.

This integrated approach addresses key limitations of traditional vocational training systems and provides a more structured learning experience.

3.3.2 Comparison with Traditional Methods

Compared to traditional vocational training approaches, the proposed system offers improved consistency, personalization, and scalability. Traditional methods rely on manual observation and instructor-led training, which can vary in quality and effectiveness.

In contrast, the system standardizes training through structured simulations and automated feedback. This ensures that all students receive consistent guidance regardless of external factors.

3.3.3 Advantages of the Proposed System

The system offers several advantages, including:

- Data-driven career recommendations
- Interactive simulation-based training
- Improved student engagement
- Scalable and cost-effective implementation

These advantages make the system suitable for deployment in special education environments.

3.3.4 Limitations and Challenges

Despite its strengths, the system has certain limitations. The dataset size is relatively small, which may affect model generalization. Additionally, some students required assistance when interacting with the system, indicating the need for further interface improvements.

The system is also limited to three job roles, which can be expanded in future versions.

3.3.5 Future Improvements

Future enhancements may include:

- Expanding the dataset for better accuracy
- Introducing adaptive difficulty in simulations
- Adding more job roles
- Improving accessibility features

These improvements will enhance system performance and usability.

3.4 Summary of Each Student's Contribution

3.4.1 Data Collection and Dataset Preparation

One member of the team was responsible for designing the questionnaire, collecting responses from students at the MJF Charitable Foundation, and preparing the dataset for analysis. This contribution ensured that the system is built on reliable and real-world data.

3.4.2 Machine Learning Development

Another member focused on developing the machine learning model, including data preprocessing, feature engineering, and model training. This role was essential in implementing the job role recommendation system.

3.4.3 Frontend Development

The frontend development was handled by a team member responsible for designing the user interface of the questionnaire and simulation modules. The focus was on accessibility and ease of use for students with cognitive challenges.

3.4.4 Backend Development and Integration

Another member worked on backend development, including API creation, database management, and system integration. This ensured smooth communication between system components.

3.4.5 Testing and Documentation

All team members contributed to system testing, validation, and documentation. This collaborative effort ensured that the system meets both technical and user requirements.

3.5 Overall Summary of Results and Discussion

The results and discussion of this study demonstrate that the Career Practice Module is an effective and practical solution for enhancing vocational training and career guidance for students with Down syndrome. The system successfully integrates multiple components, including behavioral assessment, machine learning-based prediction, and simulation-driven learning, into a unified framework designed to support personalized vocational development.

Across all evaluation phases, the system consistently showed the ability to analyze student behavioral patterns and translate them into meaningful vocational recommendations. The questionnaire-based assessment provided a structured method for capturing individual preferences, abilities, and learning characteristics. These inputs were effectively processed by the machine learning model, which demonstrated reliable performance in classifying students into appropriate job roles such as Chef Assistant, Retail Assistant, and Cleaning Assistant. The

results indicate that the system is capable of reducing subjectivity in career decision-making and replacing it with a more data-driven and objective approach.

In addition, the simulation-based learning component played a significant role in improving student engagement and skill development. By allowing learners to practice real-world job tasks in a controlled and supportive environment, the system helped bridge the gap between theoretical assessment and practical readiness. Students were able to gain hands-on experience in performing job-related activities, which contributed to improved confidence, task understanding, and learning retention. This highlights the importance of experiential learning in vocational training for individuals with cognitive and developmental disabilities.

The collaboration with the MJF Charitable Foundation significantly strengthened the validity and reliability of the study. The use of real-world data collected from students and the involvement of educators and therapists in the evaluation process ensured that the system was tested under authentic conditions. This real-world validation confirmed that the system's recommendations are not only theoretically sound but also practically applicable in actual educational environments.

Furthermore, the findings demonstrate that the integration of machine learning with vocational education has strong potential to enhance decision-making processes in special education. The system's ability to identify suitable job roles based on behavioral attributes supports the development of personalized career pathways. This is particularly important in ensuring that students are guided toward roles that align with their strengths, thereby improving long-term outcomes such as employability, independence, and social inclusion.

Despite these positive outcomes, the study also acknowledges certain limitations. These include the relatively small dataset size, the limited number of job roles, and the need for further improvements in interface accessibility and simulation adaptability. However, these limitations do not undermine the overall effectiveness of the system; rather, they provide clear directions for future enhancement and expansion.

Overall, the Career Practice Module demonstrates strong potential as a scalable and impactful solution for vocational training in special education. By combining behavioral analysis, intelligent prediction, and interactive learning, the system provides a comprehensive approach to career development. The results indicate that such technology-driven solutions can play a significant role in improving the quality of vocational education and supporting the long-term independence of students with Down syndrome.

CONCLUSION

This research presented the design and development of a Career Practice Module aimed at supporting vocational training and career guidance for students with Down syndrome. The study addressed a critical gap in existing special education systems by proposing an integrated, data-driven framework that combines behavioral assessment, machine learning-based recommendation, and simulation-based training within a single platform.

The findings of this research demonstrate that traditional vocational training approaches, which rely heavily on generalized instruction and subjective evaluation, are insufficient to meet the diverse needs of neurodiverse learners. In contrast, the proposed system introduces a structured and personalized approach that enables accurate identification of student abilities and preferences. By transforming questionnaire responses into quantifiable features, the system effectively utilizes machine learning techniques to recommend suitable job roles such as Chef Assistant, Retail Assistant, and Cleaning Assistant.

One of the key contributions of this study is the development of a real-world dataset collected in collaboration with the MJF Charitable Foundation. The involvement of teachers and therapists ensured that the dataset reflects authentic behavioral patterns, thereby improving the reliability of the machine learning model. The evaluation results indicate that the system is capable of producing recommendations that are consistent with expert assessments, highlighting its practical applicability.

Furthermore, the integration of simulation-based learning environments significantly enhances student engagement and skill acquisition. The ability to practice job-related tasks in a controlled and guided environment allows students to develop confidence and improve task performance before entering real-world workplaces. This approach effectively bridges the gap between theoretical assessment and practical readiness, which is often lacking in traditional vocational training systems.

Despite its strengths, the study acknowledges certain limitations, including the relatively small dataset size and the restriction to a limited number of job roles. These limitations suggest opportunities for future work, such as expanding the dataset, incorporating additional vocational categories, and enhancing the adaptability of simulation environments.

Overall, the Career Practice Module demonstrates strong potential as a scalable and impactful solution for inclusive vocational training. By leveraging data-driven methodologies and user-centered design, the system contributes to improving independence, employability, and social inclusion for students with Down syndrome. The outcomes of this research highlight the importance of integrating technology with special education practices and provide a foundation for future advancements in intelligent assistive learning systems.

REFERENCES

[1] J. Smith and L. Brown, "Adaptive learning strategies for individuals with Down syndrome," *Journal of Special Education Technology*, vol. 37, no. 2, pp. 85–98, 2022.

[2] Y. Lee, K. Park, and H. Kim, "Cognitive and behavioral learning patterns in students with Down syndrome," *International Journal of Developmental Disabilities*, vol. 69, no. 4, pp. 456–468, 2023.

[3] M. Garcia, R. Lopez, and P. Hernandez, "Impact of vocational training on employment outcomes for individuals with intellectual disabilities," *Disability and Rehabilitation*, vol. 44, no. 10, pp. 2145–2153, 2022.

[4] D. Thompson and S. Miller, "Enhancing workplace readiness through structured vocational programs," *Journal of Vocational Rehabilitation*, vol. 58, no. 1, pp. 23–35, 2023.

[5] UNESCO, "Inclusive education and employment strategies," UNESCO Report, 2023.

[6] R. Anderson, T. White, and K. Johnson, "Challenges in traditional vocational education for neurodiverse learners," *Education and Training in Autism and Developmental Disabilities*, vol. 58, no. 2, pp. 150–162, 2023.

[7] S. Kumar and A. Patel, "Artificial intelligence in personalized education systems: A review," *IEEE Access*, vol. 10, pp. 115234–115248, 2022.

[8] L. Chen, M. Wang, and J. Zhao, "Simulation-based learning for students with intellectual disabilities," *Computers & Education*, vol. 182, 2022.

[9] P. Silva and N. Fernando, "AI-driven vocational guidance systems for inclusive education," *IEEE Transactions on Learning Technologies*, vol. 16, no. 3, pp. 345–356, 2023.

[10] S. Kumar and A. Patel, "Artificial intelligence in personalized education systems: A review," *IEEE Access*, vol. 10, pp. 115234–115248, 2022.

[11] T. Nguyen and L. Gardner, "Machine learning approaches for personalized learning and career recommendation systems," *IEEE Access*, vol. 11, pp. 55678–55692, 2023.

[12] M. Alshammari and H. Alshammari, "Data quality challenges in machine learning for special education," *International Journal of Advanced Computer Science*, vol. 13, no. 2, pp. 88–97, 2022.

[13] R. L. Schalock and S. A. Luckasson, "Person-centered approaches and integration of expert knowledge in disability support systems," *Intellectual and Developmental Disabilities*, vol. 61, no. 2, pp. 95–108, 2023.

[14] L. Chen and X. Zhao, "Simulation-based learning for individuals with intellectual disabilities," *Computers & Education: Artificial Intelligence*, vol. 4, 2023.

[15] J. Holmes, M. Bialik, and C. Fadel, "Artificial intelligence in education: Real-time analytics and adaptive learning," *Educational Technology Research*, vol. 70, no. 3, pp. 1125–1142, 2022.

[16] P. Silva and N. Fernando, "Integrated learning systems for personalized education," *IEEE Transactions on Learning Technologies*, vol. 16, no. 3, pp. 345–356, 2023.

[17] H. Alqahtani and S. Orji, "Usability and accessibility challenges in assistive technologies," *ACM Transactions on Accessible Computing*, vol. 15, no. 2, pp. 1–25, 2023.

[18] K. Fernando and S. Perera, "Special education challenges in Sri Lanka," *Asian Journal of Inclusive Education*, vol. 10, no. 2, pp. 67–79, 2022.

[19] A. Verma and R. Singh, "AI applications in special education: Opportunities and limitations," *IEEE Access*, vol. 11, pp. 88901–88915, 2023.

[20] A. J. Carter, M. Austin, and L. Trainor, "Predictors of postschool employment outcomes for young adults with severe disabilities," *Journal of Disability Policy Studies*, vol. 33, no. 1, pp. 3–14, 2022.

[21] R. L. Schalock and S. A. Luckasson, "Enhancing person-centered planning and individualized supports for people with intellectual disabilities," *Intellectual and Developmental Disabilities*, vol. 61, no. 2, pp. 95–108, 2023.

[22] E. S. Wehman, K. Chan, and M. Brooke, "Transition from school to work for individuals with intellectual disabilities: Current challenges and future directions," *Journal of Vocational Rehabilitation*, vol. 58, no. 2, pp. 123–135, 2023.

[23] P. J. Brown and D. H. Smith, "Limitations of traditional vocational training models in special education," *International Journal of Special Education*, vol. 37, no. 1, pp. 45–58, 2022.

[24] S. K. Rao and M. Meo, "Subjectivity in educational assessment and its impact on learner outcomes," *Educational Assessment Review*, vol. 34, no. 3, pp. 210–224, 2022.

[25] T. Nguyen and L. Gardner, "Machine learning approaches for personalized learning and career recommendation systems," *IEEE Access*, vol. 11, pp. 55678–55692, 2023.

[26] H. Alqahtani and S. Orji, "Usability and accessibility challenges in assistive technologies for individuals with cognitive disabilities," *ACM Transactions on Accessible Computing*, vol. 15, no. 2, pp. 1–25, 2023.

[27] J. M. Ruppar, R. Alper, and M. Gaffney, "Bridging the gap between school and employment for students with disabilities," *Career Development and Transition for Exceptional Individuals*, vol. 46, no. 1, pp. 12–24, 2023.

[28] L. Chen and X. Zhao, "Simulation-based vocational training systems for individuals with intellectual disabilities," *Computers & Education: Artificial Intelligence*, vol. 4, 2023.

[29] K. Fernando and S. Perera, "Challenges in special education and vocational training in Sri Lanka," *Asian Journal of Inclusive Education*, vol. 10, no. 2, pp. 67–79, 2022.